Temporal Complexity Analysis (USEI09)

Algorithm: k-Nearest Neighbors (k-NN) on 2D-Tree with Filtering

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1. Complexity Overview

The complexity of the k-NN search depends heavily on the efficiency of the **Pruning Strategy**. The algorithm attempts to ignore subtrees that cannot possibly contain a station closer than the N-th neighbor found so far.

- **Best Average Case: O(log n)**

- **Condition:** N is small relative to n, and the optional Filter matches stations in the geometric vicinity of the target.
- **Example:** Finding the 5 nearest stations to a coordinate in Paris (with no filter or a filter for "FR").
- **Justification:** The algorithm quickly locates valid neighbors. The distance to the N-th neighbor becomes a tight **"pruning threshold."** As this threshold shrinks, the algorithm discards large sections of the tree (entire subtrees) because their bounding boxes (hyper-rectangles) are farther away than the neighbors already found.

- **Worst Case: O(nlog N)**

- **Condition:**

    1. **Strict Filtering:** Searching for stations with TimeZone="Asia/Tokyo" while targeting coordinates in Portugal. The algorithm visits local nodes but rejects them due to the filter. It cannot shrink the pruning threshold and is forced to expand the search radius until it traverses the entire tree to find valid matches.
    2. **Large N:** Requesting N approx n (total stations).

- **Justification:** The "pruning check" (distToSplittingPlane < maxDist) almost always evaluates to true, forcing the recursion to visit both the Left and Right children at every node. The log N factor comes from the cost of inserting/removing elements from the Priority Queue (Max-Heap) for every visited node.

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2. Component Costs

The total cost is derived from the following operations:

Component	Cost	Description
Tree Traversal	O(log n) to O(n)	Depends on pruning efficiency.
Priority Queue	O(log N)	Cost to insert a new result or poll the farthest one.
Distance Calc	O(1	Haversine formula calculation.
Filter Check	O(1)	HashSet lookup (Time Zone Group check).

Definitions:

- n: Total number of stations (approx 64,000).
- N: The desired number of neighbors (k).

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3. Speedup vs. Linear Scan

- **Linear Scan:** Always requires O(n) iterations (calculating distance to every single station).
- **2D-Tree:** Varies dynamically.
    - **Performance Note:** For typical operational queries (e.g., N=10), the 2D-Tree approach is approximately **100x to 500x faster** than a linear scan, provided the filter does not force a full tree traversal.